DEEP LEARNING LABORATORY

EXPERIMENT 7

PERCEPTRON AND FORWARD PROPAGATION

Aim

To implement a perceptron for the AND logic gate and observe forward propagation by calculating the weighted sum of inputs, applying a threshold activation function, and comparing predicted outputs with target values.

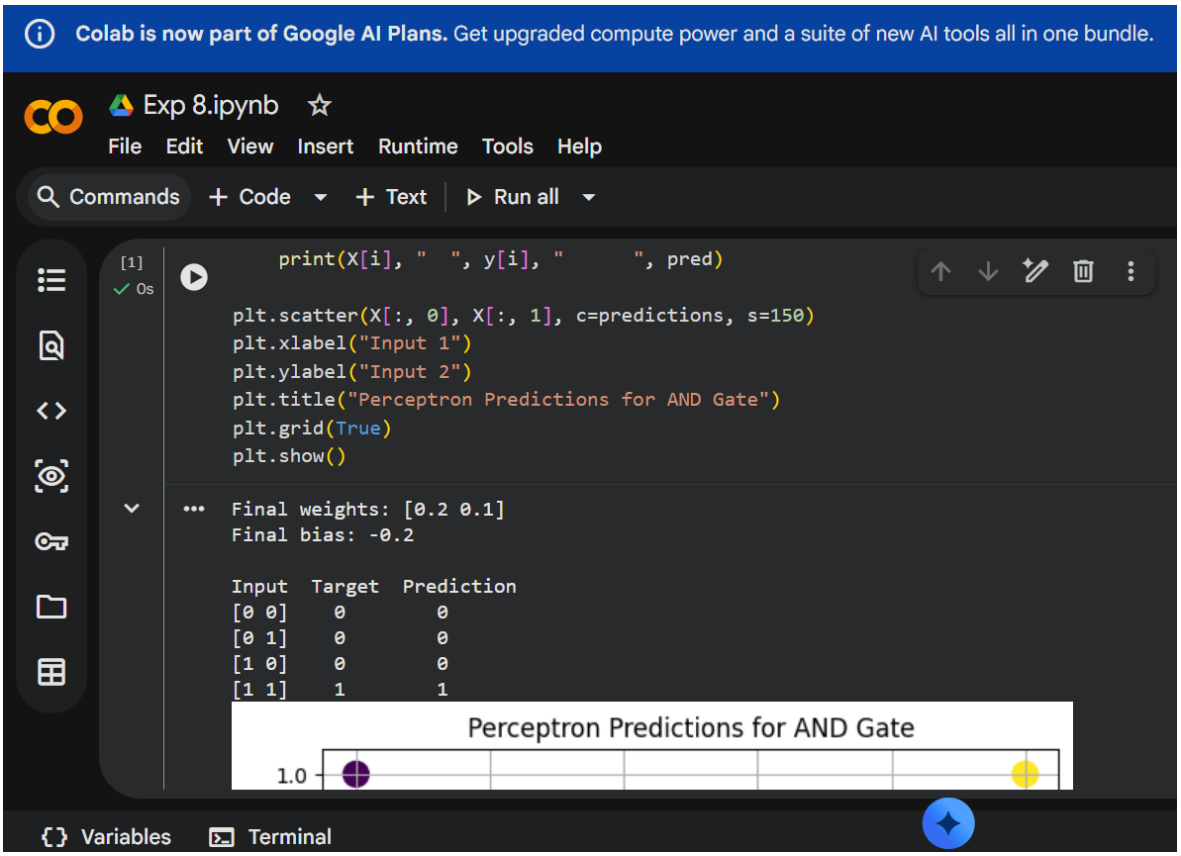
Output

The perceptron learns the AND gate correctly. The final weights are [0.2, 0.1] and the bias is –0.2. The predictions match all four target outputs.

Observed output table

Input	Target	Prediction
[0, 0]	0	0
[0, 1]	0	0
[1, 0]	0	0
[1, 1]	1	1

Colab output screenshot



Prediction graph

